



LJ University
University with a Difference

**LJ INNOVATION
VILLAGE**

MEET THE TEAM



PRIYANIRATHOD
SEM - 1 : BE - AIDS



PUSHTI TRAMBADIYA
SEM - 1 : BE - IT



RAVI TRIVEDI
SEM - 1 : BE-IT



HET VEDANI
SEM - 1 : BE - IT



SURYANSH CHANDAK
SEM - 1 : BE -AIDS



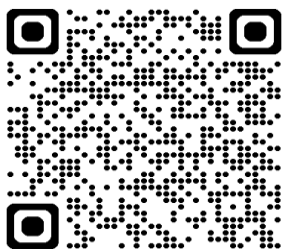
LET'S CONNECT

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LJ INSTITUTE OF ENGINEERING
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FEEL TO READ: A BRAILLE COMPANION

“Empowering Learning Through Touch.”

The project is an Arduino-based Braille learning system that converts user input into tactile Braille patterns using solenoids. It applies electromagnetism and embedded system concepts to enable quiet, low-cost, and independent learning for visually impaired users, making it suitable for inclusive educational environments

01 WHY ARE WE DOING THIS PROJECT? (PROJECT RATIONALE).

- Visually impaired students face major difficulty accessing printed learning materials.
- Braille books are expensive, bulky, and slow to produce.
- Our project provides a low-cost, real-time solution that converts normal text into tactile Braille, enabling independent learning and improved accessibility.

02 WHICH PROBLEM DOES IT ADDRESS?

- Many visually impaired learners cannot read printed text without assistance.
- Existing Braille materials require long preparation time and are not always available.
- This creates dependency on others and limits educational opportunities.
- Our system helps by instantly converting digital text into physical Braille dots.

03 WHAT ARE THE ALTERNATE SOLUTIONS OUT THERE?

- Current alternatives include printed Braille textbooks, refresh a Braille displays, and audio-based screen readers.
- While effective these solutions are often costly, complex, and not affordable for many institutions or individuals.
- Some systems are bulky and require specialized infrastructure.

04 WHAT MAKES OUR PROJECT STAND OUT FROM THE REST? THIS IS YOUR UNIQUENESS OR INNOVATION.

- Our project is unique because it is affordable, simple, and suitable for education.
- It generates Braille in real time using easily available components independent of dot control, demonstrates embedded systems and electromagnetism, and can be deployed in schools and colleges at much lower cost.